SKYLAR

skylar.rocks | github.com/SkylarS300

EDUCATION

Washington University in St. Louis — B.S. Biomedical Engineering, McKelvey School of Engineering

John B. Ervin Scholar • Minor in Writing • Fall 2026 – Present

High School for Environmental Studies, New York, NY

Graduated June 2026

Carnegie Mellon University — Pre-College Program in Computational Biology

Summer 2025 • Selected as one of 97 students from 400+ applicants

- Conducted wet-lab work: microbiome sampling, DNA extraction, PCR, gel electrophoresis, Sanger sequencing, antibiotic resistance assays
- Implemented computational pipelines: sequence alignment, genome assembly, phylogenetic tree inference, SARS-CoV-2 variant tracking

RESEARCH & BIOINFORMATICS PROJECTS

Independent Research Portfolio (2024 – Present)

Original computational biology analyses using real RNA-seq data

- **Tumor vs. Fetal Mitosis:** compared ~9,000 TCGA tumor samples with fetal RNA-seq (597 genes × 120 samples) using PCA, Euclidean distance matching, and clustering; found tumor–fetal similarity across proliferative tissues (liver, heart, kidney); identified marker genes MKI67 and PLK1
- **Circadian Gene Dysregulation in Pancreatic Cancer:** analyzed the TCGA-PAAD dataset for 32 circadian regulators; identified key dysregulation patterns (RAB24↓, WDR75↑); volcano and box plots for differential expression
- **Breast Cancer Gene Expression Subtypes:** processed TCGA-BRCA RNA-seq counts with EdgeR and limma-voom; GO/KEGG enrichment revealed mitotic, lipid, and ECM remodeling pathways across subtypes

Transcriptome Explorer — Bioinformatics Web App

React + FastAPI

- Cross-species RNA-seq analysis platform: PCA, differential expression, volcano plots, enrichment, CSV export; UMAP visualization and persistent sessions in development

OpenBio Tools — Creator & Developer

React + FastAPI

- Modular research toolkit: Protein Interaction Explorer, ORF Translator, RNA-Seq Visualizer, and CRISPR Guide Picker, designed for accessible student-friendly bioinformatics

MAJOR PROJECTS

LearnLoom — Literacy Web App

Full-stack developer

- Reading comprehension and grammar platform with text-to-speech read-aloud, grammar quizzes, and accessibility tools; presented to PwC as an equitable ed-tech solution and piloted in classrooms

Miranda — AI/ML Legal Rights Chatbot

Kode With Klossy, AI/ML track

- Hybrid NLP chatbot (React + Gradio) for plain-language legal rights education; presented at Apple Fifth Avenue and deployed on Hugging Face

OurChoice — Civic Data Visualization Project

Kode With Klossy, Data Science track

- 100+ visualizations on post-Roe abortion access and maternal mortality trends using ArcGIS, Seaborn, and Matplotlib; 10,000+ online viewers

Human Futures — Civic Learning Platform

Collaboration with Rep. Jamaal Bowman's team

- Advised on UX design, GitHub integration, and interactivity; proposed AI-assisted educator tools and a mission builder for classrooms

LEADERSHIP & ACTIVITIES

- **Peer Tutoring Club** — President: lead weekly tutoring in biology, chemistry, and English; built systems to assign tutors and track outcomes
- **Coding Club** — President: teach Python, JavaScript, HTML/CSS, and Git; guide peers in building personal portfolios and projects
- **Business Club** — Vice President: lead workshops on marketing, personal branding, and financial literacy; coach peers on pitching and public speaking
- **Student Council** — Vice President: represented the student body in schoolwide decision-making and organized campaigns
- **National Honor Society** — Community Relations Officer: liaison for academic outreach and accessibility projects
- **Robotics Club** — Lead Programmer: programmed autonomous and driver-controlled systems in Python/Java

EXPERIENCE

Student Intern, theCoderSchool

Aug 2025 – Present

- Mentor beginner and intermediate coders in Python and JavaScript; facilitate team projects and inclusive coding environments

Private Chemistry Tutor

Jan – May 2025

- Designed custom review materials and practice sets; improved student mastery of stoichiometry, kinetics, and equilibrium

Volunteer

- Discovery Guide, Central Park Zoo • Voter Registration Organizer • Phone Bank Volunteer, Arizona Midterms

AWARDS & HONORS

- John B. Ervin Scholars Program, Washington University in St. Louis (2026)
- Scholastic Art & Writing Awards — National Silver Medal, Portfolio; Louise DiMichele Scholarship (2026)
- Scholastic Art & Writing Awards — Regional Gold Key and Silver Key, Portfolio (2026)
- Posse Finalist, Smith College (2025)
- Winner, theCoderSchool National Python Contest (2025)
- College Board National Recognition Program: Top School Honoree (2025)

SKILLS

Technical: Python (Pandas, SciPy, scikit-learn, Seaborn, Matplotlib, Plotly), JavaScript, React, FastAPI, SQL, HTML/CSS, Tailwind, Git/GitHub

Bioinformatics: RNA-seq analysis, PCA, differential expression, functional enrichment (g:Profiler), reproducible workflows

Leadership & Communication: public speaker; presented technical and civic projects to PwC, Apple, and congressional staff